



Indian Institute of Technology Guwahati

BSc. (Hons.) Data Science & Artificial Intelligence
Online Degree Programme

INTERNSHIP

InsightPulse: AI-Powered Multi-Channel Survey & Unified Analytics Platform

Student Name :	Bendi Karthikeya
Student ID :	23035010310
Email ID :	k.bendi@op.iitg.ac.in
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Trimester :	IX

Introduction & Internship

- **Who I am.**

Bendi Karthikeya, BSc (Hons) Data Science & AI, IIT Guwahati (Online). This is my Internship-II (DA 378) submission.

- **Where I intern.**

Full Stack Generative AI Intern at FIT - FIRST Information Technology (Muscat, Oman; remote), building production-grade applications for the company's clients.

- **This project.**

InsightPulse is the client project I built at FIT: an AI-powered multi-channel survey platform collecting feedback via Voice (VAPI), WhatsApp, Web, and Email with real-time unified analytics.

The Problem & Why It Matters

- **Feedback collection is fragmented.**

Surveys are distributed across separate, disconnected systems (email, SMS, web, voice) with no unified data structure or central storage.

- **Voice & WhatsApp data is lost.**

Voice conversations are not transcribed or analyzed, and messaging is managed manually, leading to delayed feedback loops.

- **Analysis lacks depth and speed.**

Manual categorization of user comments is slow, subjective, and unable to extract emotional context or action items in real-time.

- **The goal.**

A unified platform where web, WhatsApp, and AI voice surveys dump into one database, with real-time analytics and LLM-powered emotion scoring.

Tools & Technologies

This project, InsightPulse

- React 18 · TypeScript (Frontend layout)
- Node.js · Express (Backend API server)
- Neon Serverless PostgreSQL (Database storage)
- Drizzle ORM (Database query schema)
- VAPI Voice AI & Meta WhatsApp Cloud API
- OpenRouter (DeepSeek/Gemini fallback APIs)

Core Features Highlight

- Real-time TanStack Query dashboard (30s poll)
- Plutchik's 8-Emotion LLM scoring model
- Zoho CRM OAuth contacts integration
- Dynamic question builder & localized translation
- Automated department NPS scoring & action plans

System Ingested Input

- **Real-time feedback stream.**

The system does not process static datasets, but handles real-time feedback submissions from active channels.

- **Quantitative & Qualitative inputs.**

Captures both numerical scores (CSAT 1-5, NPS 0-10, EVI 0-100) and open-ended customer comments.

- **Asynchronous normalization.**

Numerical feedback and raw transcripts are structured dynamically. Raw texts are queued for OpenRouter emotion extraction, then saved in SQL.

How It Works: Data Pipeline

Multi-channel ingestion with unified analytical storage

Feedback flows from channels -> webhook / API -> OpenRouter analysis -> DB save -> TanStack Query UI updates.

AI & Voice AI Integration

Voice calls via VAPI, text analysis via OpenRouter (DeepSeek/Gemini failover)

- **Structured VAPI Callbacks.**

VAPI Voice assistant calls customers, asks localized (EN/AR) questions, captures voice responses, extracts structured survey scores, and calls the webhook.

- **OpenRouter LLM Pipeline.**

Web and WhatsApp survey comments are sent to OpenRouter. Strict system prompts enforce JSON mode and a temperature of 0.3 ensures consistent evaluation.

- **Sentiment & Plutchik Emotions.**

LLM rates 8 Plutchik emotions (0-10), extracts sentiment, EVI score, category/theme, and generates real-time recommendations and action items.

- **Resilient Failover Chain.**

Primary model is DeepSeek-Chat. If the API fails or returns invalid JSON, it transparently falls back to Gemini-2.0-Flash on OpenRouter.

Where I Learned This

Course: Large Language Model Application Engineering (DA 3022)

Large Language Model Application Engineering

Course DA 3022

- Prompt engineering
- Structured outputs
- Chain-of-thought
- Guardrails & reliability

Concept → where it is in my code

- **Prompt anatomy**

role + context + 8-emotion rubric + instructions + JSON schema →
buildAnalysisPrompt()

- **Chain-of-thought**

the STEP 1 parse -> 2 categorise -> 3 evaluate emotions -> 4 recommend
actions protocol

- **Structured output / JSON mode**

response_format = {"type": "json_object"} + Zod validation schema

- **Reliability over creativity**

temperature 0.3 for reproducible sentiment scores

- **Guardrails**

'Return ONLY valid JSON', 'do not add markdown formatting or extra text'

CRM Integration & Custom Flows

Zoho CRM OAuth, WhatsApp Sequences & Dept NPS

- **Zoho CRM Integration.**

OAuth 2.0 flow allows administrators to fetch client contacts from Zoho, trigger multi-channel surveys, and sync responses back to CRM records.

- **WhatsApp Session State.**

WatiSessions tracking allows sequential question delivery (checking response data types per step) to avoid multi-message clutter on WhatsApp.

- **Department-wise NPS.**

Aggregates Net Promoter Score dynamically for departments, mapping employee feedback back to their respective business units.

- **Scheduled Survey Cron.**

Custom survey scheduler executes in the background to automatically dispatch pending survey campaigns via email/WhatsApp.

Outputs & Results

Demonstrated in the video walk-through of the live platform

- **Trigger Outbound Voice AI.**
Initiating a VAPI call dials the target, conducts the survey, and displays the transcript and structured output in the dashboard.
- **Unified Real-Time Dashboard.**
React UI displays EVI, NPS, CSAT, and CES scores with period-over-period trends and 30s auto-refresh.
- **LLM Emotion Radar.**
Sentiment classification and Plutchik's 8-emotion distribution charts dynamically render customer emotional profiles.
- **Zoho CRM Synchronization.**
Surveys triggered for CRM contacts correctly update fields, enabling automated customer relationship management loops.

9 · Wrap Up

I built InsightPulse at FIT: a multi-channel feedback system incorporating Voice AI and WhatsApp to centralize customer sentiment into a single analytics platform.

Biggest takeaway: applying prompt engineering & structured JSON output from course DA 3022 to handle complex, multi-channel inputs and model failover configurations.

Future work: implementing LangGraph for conversational multi-turn survey follow-ups, semantic clustering of user complaints, and real-time negative-sentiment email routing.

Code Repository: <https://github.com/workingaditya/insightpulse>

Video Demo: <https://insightpulse.skill-pulse.io>

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